

MATHEMATICS SUBJECTIVE TEST

Class X (Session 2026 – 27)

Duration : 120 Minutes

Max. Marks : 45

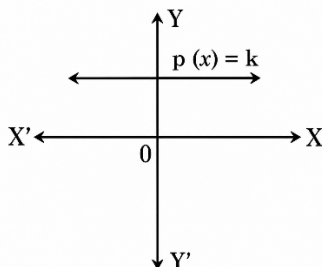
General Instruction :

1. All questions are compulsory.
2. Section A – 5 questions $\times$ 1 mark = 5 marks.
3. Section B – 5 questions $\times$ 3 marks = 15 marks.
4. Section C – 5 questions $\times$ 5 marks = 25 marks.

SECTION – A

(Each question carries 1 mark)

1. The equation $x + \frac{1}{x} = 3$ ($x \neq 0$) is expressed as a quadratic equation in the form $ax^2 + bx + c = 0$. Find the value of $a - b + c$. (1 marks)
2. **Assertion (A) :** The value of p for which the system of equations $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ is consistent is 4. (1 marks)
Reason (R) : The system of equations $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$ is consistent with infinitely many solutions, if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.
 - (a) Both true, Reason is correct explanation
 - (b) Both true, Reason is not the correct explanation
 - (c) Assertion is true, Reason is false
 - (d) Assertion is false, but Reason is true.
3. The graph of polynomial $p(x) = k$ is shown here. Number of zeroes of polynomial $p(x)$ is : (1 marks)



4. If $(-1)^n + (-1)^8 = 0$, then what is n . (1 marks)
5. Solve the quadratic equation : $x^2 - 2ax + (a^2 - b^2) = 0$ for x . (1 marks)

SECTION - B

(Each question carries 3 mark)

6. In the picture given below, one can see a rectangular in-ground swimming pool installed by a family in their backyard. There is concrete sidewalk around the pool that is width x m. The outside edges of the sidewalk measure 7m and 12m. The area of the pool is 36 sq.m.



Based on the information given above, form a quadratic equation in terms of x . Find the width of the sidewalk around the pool. **(3 marks)**

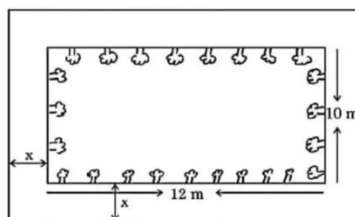
7. If the sum of the roots of the quadratic equation $ky^2 - 11y + (k - 23) = 0$ is $13/21$ more than the product of the roots, then find the value of k . **(3 marks)**
8. If α, β are the zeroes of the quadratic polynomial $px^2 + qx + r$, then find the value of $\alpha^3\beta + \beta^3\alpha$. **(3 marks)**
9. Neha claimed that there does not exist any irrational number between 1 and 2. Raunak claimed that $\sqrt{2}$ lies between 1 and 2 and $\sqrt{2}$ is an irrational number. Who do you think is correct? Justify by proving either $\sqrt{2}$ is an irrational number or otherwise. **(3 Marks)**
10. In a teachers' workshop, the number of teachers teaching **French, Hindi, and English** are **48, 80, and 144** respectively. Find the **minimum number of rooms required** if, in each room, the **same number of teachers** are seated and **all of them are of the same subject**. **(3 Marks)**

SECTION - C

(Each question carries 5 mark)

11. A garden designer is planning a rectangular lawn that is to be surrounded by a uniform walkway.

(5 marks)



The total area of the lawn and the walkway is 360 square metres. The width of the walkway is same on all sides. The dimensions of the lawn itself are 12 metres by 10 metres. Based on the information given above, answer the following questions :

- (i) Formulate the quadratic equation representing the total area of the lawn and the walkway, taking width of walkway = x m.
- (ii) (a) Solve the quadratic equation to find the width of the walkway ' x '.

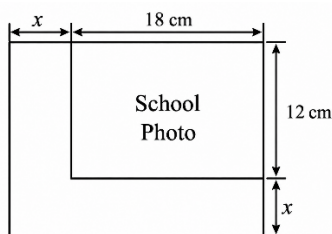
OR

- (b) If the cost of paving the walkway at the rate of Rs. 50 per square metre is Rs. 12,000, calculate the area of the walkway.
- (iii) Find the perimeter of the lawn.

12. **Case Study:** While designing the school yearbook, a teacher asked the student that the length and width of a particular photo be increased by n units each to double the area of the photo. The original photo is 18 cm long and 12 cm wide.

Based on the above information, answer the following questions :

- Write an algebraic equation depicting the above information.
- Write the corresponding quadratic equation in standard form.
- What should be the new dimensions of the enlarged photo?



OR

Can any rational value of x make the new area equal to 220cm^2 ?

(5 marks)

13. Seema daily goes to a park to exercise on machines available there. When Seema spent **15 minutes on exercise bicycle** and **30 minutes on double cross walker**, she received a message of burning **435 calories** on her fitness watch. When she spent **30 minutes on exercise bicycle** and **40 minutes on double cross walker**, she received a message of burning **690 calories**. To find the number of calories burned per minute on each machine, answer the following :

(5 Marks)



- Represent the above situation in terms of a pair of linear equations in two variables.
 - Show that the equations have a unique solution.
 - Solve both equations to find the values of the variables using **elimination method**.
 - Solve both equations to find the values of the variables using **substitution method**.
14. Vijay invested certain amounts of money in two schemes **A** and **B**, which offer interest at the rate of **8% per annum** and **9% per annum**, respectively. He received **₹1,860** as the total annual interest. However, had he interchanged the amounts of investments in the two schemes, he would have received **₹20 more** as annual interest. How much money did he invest in each scheme? (5 Marks)
15. An arch of a railway bridge, built on Chenab riverbed, is shown in the above diagram. It is a parabolic arch connecting two hills at P and Q. If the parabolic curve is represented by the polynomial $p(x) = -0.0025x^2 - 0.025x + 136$. (5 Marks)

Observe the diagram and based on above information, answer the following questions :

- Write the co-ordinates of point A.
- Find the span of the arch
- Write the zeroes of the polynomial using diagram and verify the relationship between sum of zeroes and polynomials.

(b) Find the values of $p(x)$ at $x = 100$ and $x = -100$. Are they same?

